6.4 WEIGHTED INDEX NUMBERS

In computing weighted Index Numbers, the weights are assigned to the items to bring out their economic importance. Generally quanties consumed or value are used as weights.

Weighted index numbers are also of two types

(i) Weighted aggregative

(ii) Weighted average of price relatives

6.4.1 Weighted aggregate Index Numbers

In this method price of each commodity is weighted by the quantity sale either in the base year or in the current year. There are various methods of assigning weights and thus there are many methods of constructing index numbers. Some of the important formulae used under this methods are

a) Laspeyre's Index ($P_{01}{}^{L}$)

b) Paasche's Index ($P_{01}{}^{P}$)

c) Dorbish and Bowley's Index ($P_{01}{}^{DB}$)

d) Fisher's Ideal Index ($P_{01}{}^{F}$)

e) Marshall-Edgeworth Index ($P_{01}{}^{Em}$)

f) Kelly's Index ($P_{01}{}^{K}$)

a. Laspeyre's method

The base period quantities are taken as weights. The Index is

$$P_{01}^{L} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

b. Paasche's method

The current year quantities are taken as a weight. In this method, we use continuously revised weights and thus this method is not frequently used when the number of commodities is large. The Index is

$$P_{01}^{P} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$$

c. Dorbish and Bowley's method

In order in take into account the impact of both the base and current year, we make use of simple arithmetic mean of Laspeyre's and Paasche's formula

The Index is

$$P_{01}^{DB} = \frac{P_{01}^{L} + P_{01}^{P}}{2}$$

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